Locating Mobile Station with Ellipse Propagation Model Based on GSM Networks

Presenter : Junyang Zhou
Supervisor : Dr. Joseph Kee-yin Ng
Co-Supervisor: Dr. Jianliang Xu

Content

Background

My work

Current Findings

Future work

Q&A

Background

Positioning Technology

Global Positioning System (GPS)

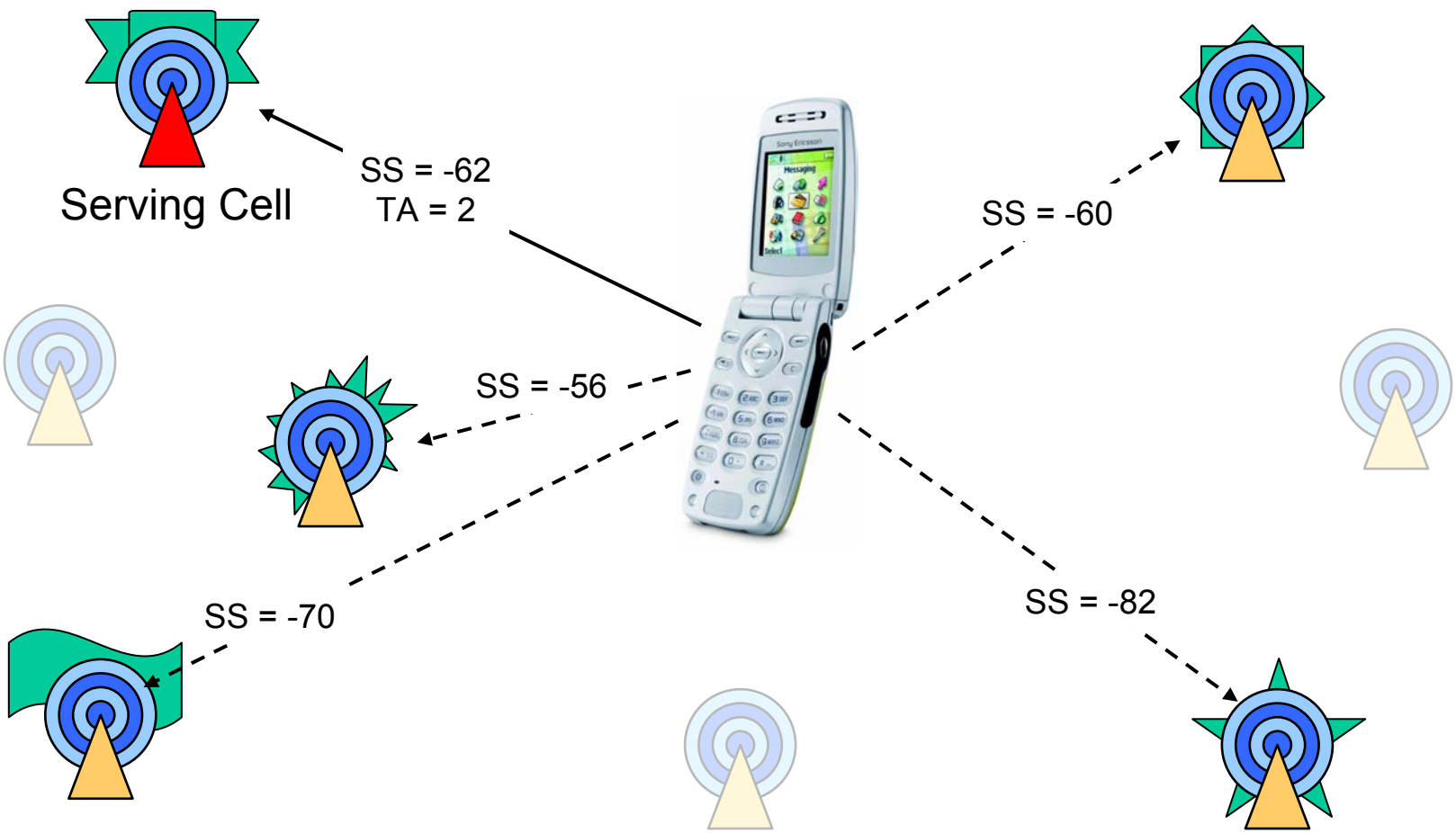
- A GPS receiver measures the distances from 3 or more satellites and utilize a mathematical method, called "Triangulation" to calculate the actual position
- Blind Spot Problem: Signals from the satellites would be blocked, and reflected by tall buildings.
- Compensations
  - Position Reckoning: Electronic Compass, and Speedometer
  - Ground Stations

Positioning Technology

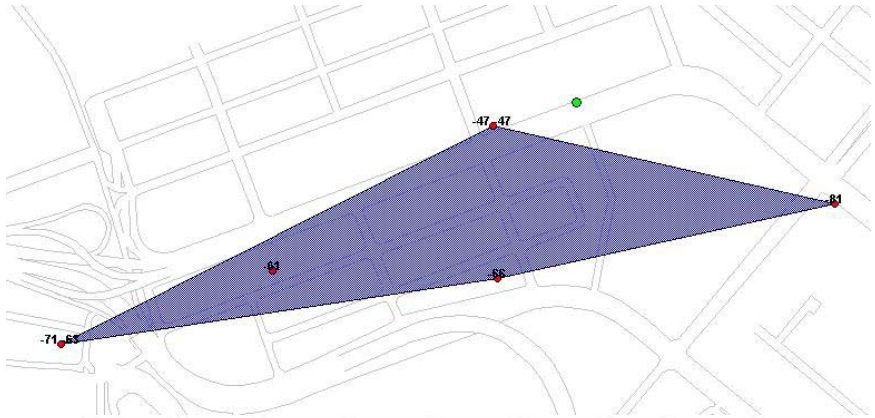
Location Estimation based on Cellular Phone Network

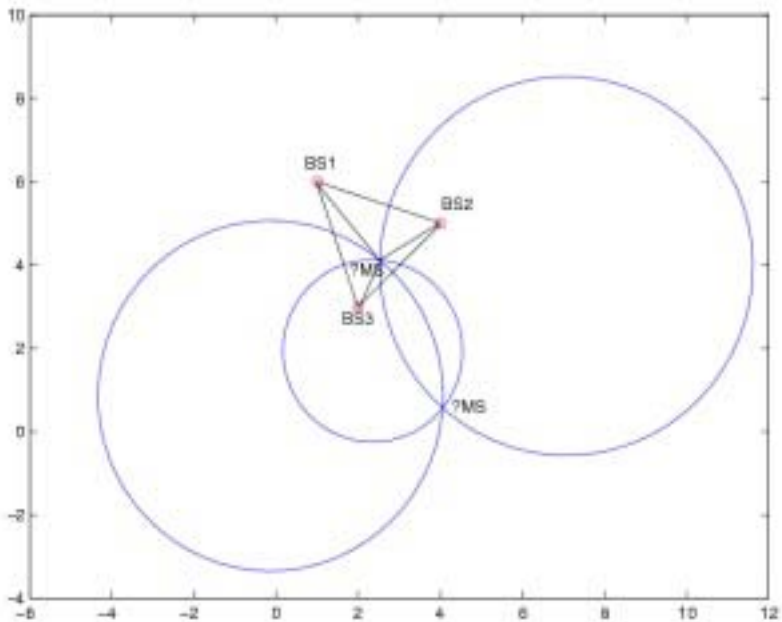
- CDMA: ToA, TDoA, E-OTD, AoA
- GSM: Signal Strength Based Algorithms
- Accuracy depends on the cell type & layout
  - Pico-cell: 10-1000m
  - Micro-cell: 100-1000m
  - Small Macro-cell: 1000-3000m
  - Large Macro-cell: 3000-30Km
- Feasible for Metropolitan areas like Hong Kong
- Compliment to GPS

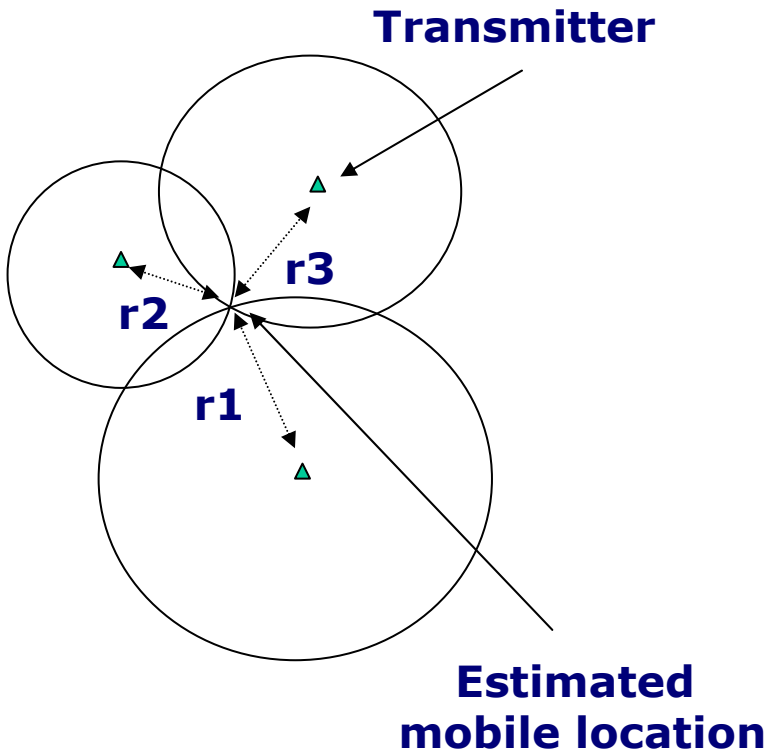
GSM Network



Signal Strength based Models: CG ,CT & Locus





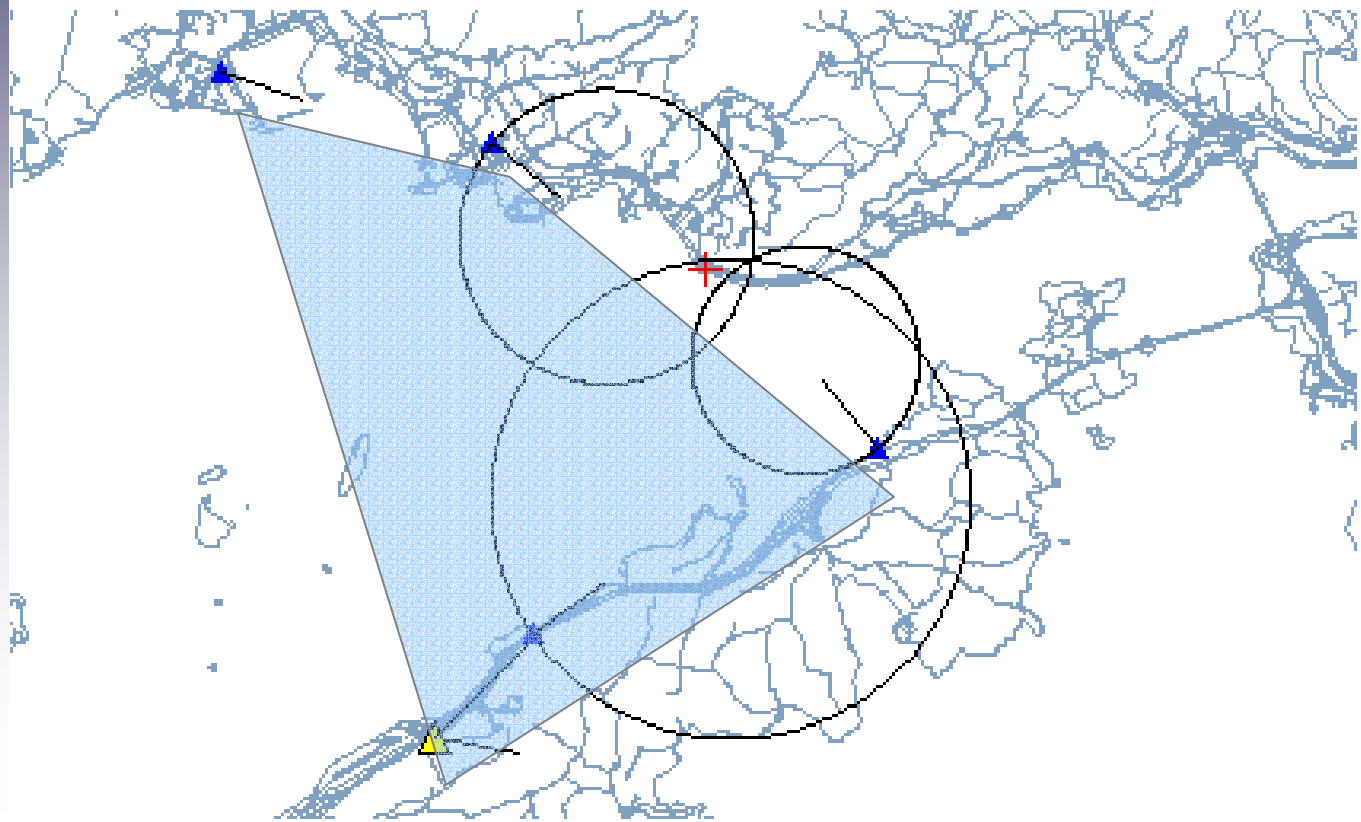


Fine Tuning
Clustering
Eliminating Bad Data

Center of Gravity(CG) vs. Circular Trilateration(CT)

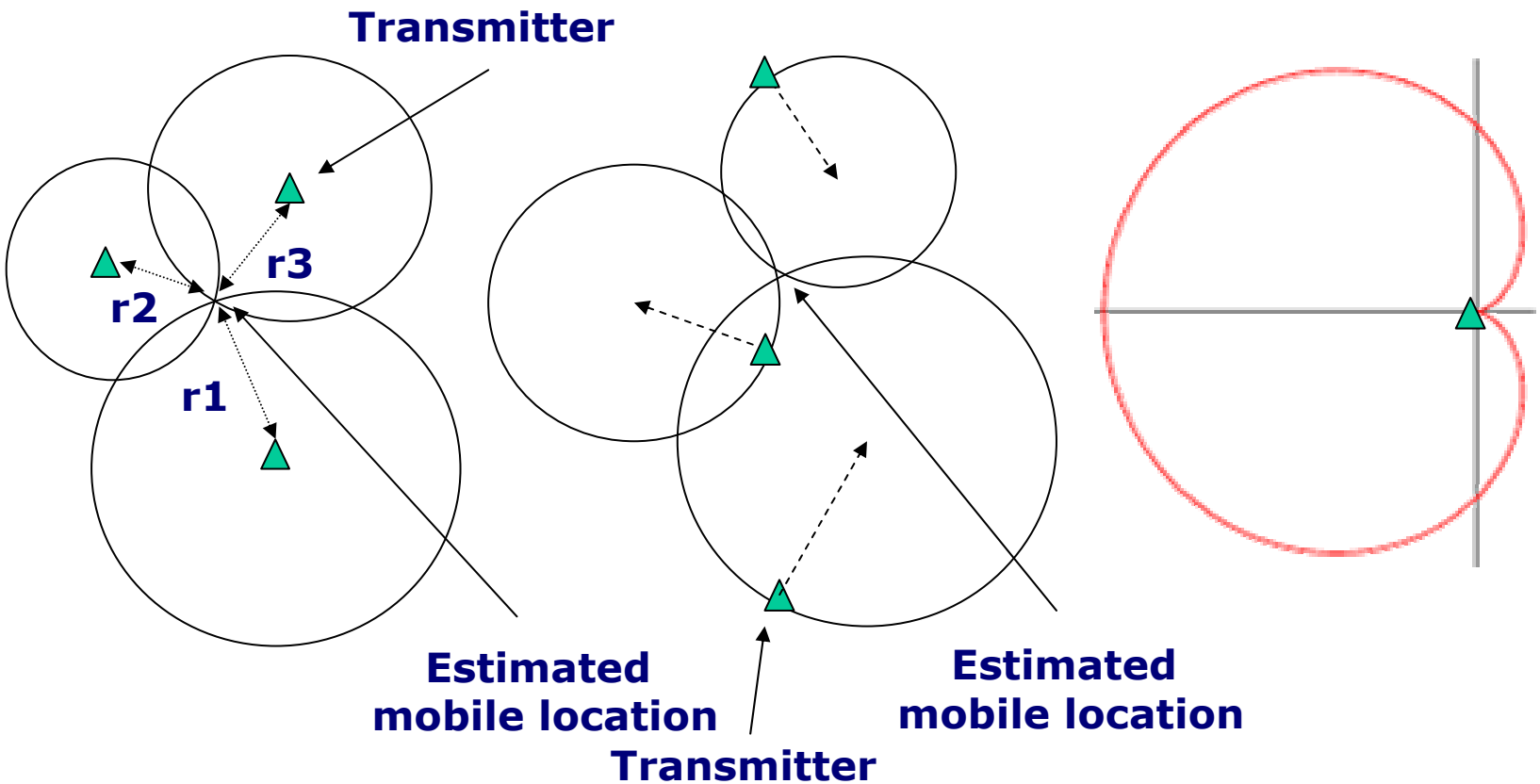
The mobile location can be outside the convex hull

CT will be suitable for seashores or some remote areas

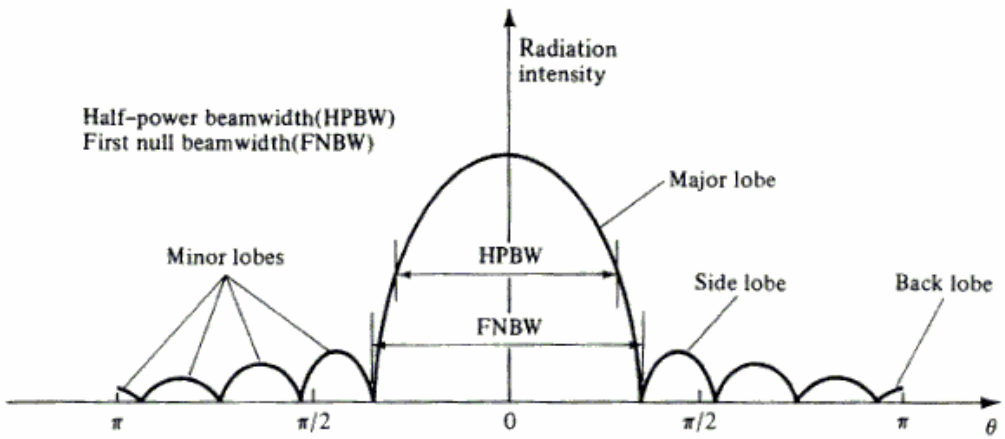


Circular Trilateration

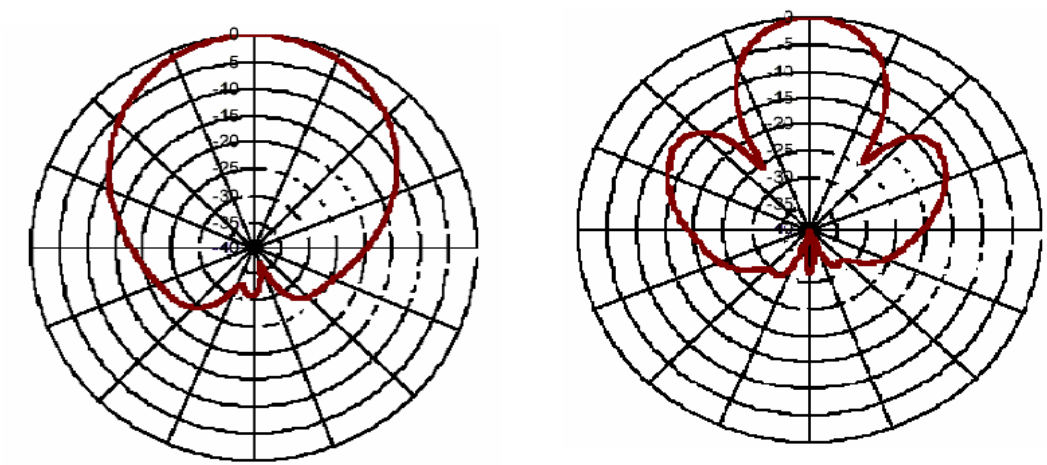
- Circle
- Circle with Bearing
- Cardioid



Directional Signal Propagation Model



Involve Directional Property of All Antenna Types



My work

Ellipse Propagation Model

Main Idea:

- Consider the effect of antenna signal directional transmission
- Suppose the contour line of signal strength is an ellipse
- Consider the base station is on one of the focus
- Modify from the original propagation model
- Present a new directional propagation model to describe the relationship between the distance and the signal strength

Ellipse Propagation Model

- Translate the signal strength into distance

The Ellipse Propagation Model :

$$d = k(s_0/s)^{1/\alpha}(1-e)/(1-e\cos(\theta))$$

Where

θ is the deviation, and s is the signal power received,
s_0 is the transmitting power of the base station,
k and α are constants,

Geometric Algorithm

Main Idea

- Use three base stations which are not in a line to locate the mobile station
- Use Ellipse Propagation Model to translate the signal strengths received from the base stations into the distances between the base stations and the mobile station
- Solve three equation groups derived from the three circles formula
- Choose the center of these three solutions as the estimation of the mobile station

Geometric Algorithm

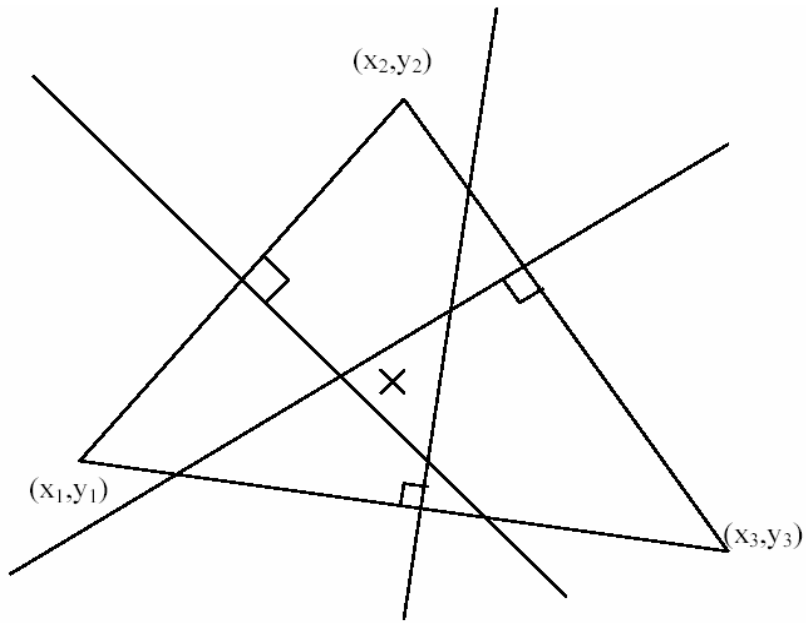
- Geometric Algorithm

$$d_1^2 = (x - x_1)^2 + (y - y_1)^2$$
$$d_2^2 = (x - x_2)^2 + (y - y_2)^2$$
$$d_3^2 = (x - x_3)^2 + (y - y_3)^2$$

- Generate three equation groups from these three circles formula
- Each equation group can only derive one solution
- Choose the mean of these three solutions as the estimation

Geometric Algorithm

- The Geometric Interpretation of Geometric Algorithm



- The three points on the triangle are the three base stations
- The “cross” point is the estimation
- Each intersection is the solution of the equation group derived from these three circles formula

Iterative Algorithm

Main Idea

- Use EPM to translate the signal strength into the distance
- Consider the real location of the mobile station as the conditional expectation of the information of signal strengths and the base station locations
- Consider the estimation of the mobile station as the certain part of the conditional expectation
- Build up the iterative formula to find out the estimation
- Choose the convergence point as the estimation of the mobile station

Iterative Algorithm

Consider the "real" location is the conditional expectation about the information we have, denoted it

$$x_1 = E(x|x_0, y_0, s, l), \text{ and } y_1 = E(y|x_0, y_0, s, l).$$

- Rewrite it as two parts: one is certain, one is uncertain.

$$x = f(x_0, y_0; s_1, s_2, ..., s_m; l_1, l_2, ..., l_m) + \epsilon,$$

$$y = g(x_0, y_0; s_1, s_2, ..., s_m; l_1, l_2, ..., l_m) + \eta.$$

Iterative Algorithm

Estimation of the Mobile Station

$$x_0 = f(x_0, y_0; s_1, s_2, ..., s_m; l_1, l_2, ..., l_m),$$

$$y_0 = g(x_0, y_0; s_1, s_2, ..., s_m; l_1, l_2, ..., l_m).$$

- x_0 and y_0 appear both side in the equations
- Solve these equations by iterative method

Iterative Algorithm

Iterative formulas

$$x_n = f(x_{n-1}, y_{n-1}; s_1, s_2, ..., s_m; l_1, l_2, ..., l_m),$$
$$y_n = g(x_{n-1}, y_{n-1}; s_1, s_2, ..., s_m; l_1, l_2, ..., l_m).$$

- Choose the Geometric Algorithms function structures as f and g structures
- Choose one point as initial value
- Under some conditions, It can be proved the above iterative formulas have convergence point. This is the estimation of the mobile station

Statistical Estimation

Main Idea

- Use more than one snapshot information to find out the estimation
- Use the Geometric Algorithm and the Iterative Algorithm based on the Ellipse Propagation Model

Statistical Estimation

Define an information set

$$\text{Set } D_t = \{x_t, y_t, D_{t-1}\}, \text{ and } D_0 = \{x_0, y_0\}$$

Derive a series of estimations from n snapshots

$$\begin{aligned}
&x_1 = E(x|D_0), y_1 = E(y|D_0);\\
&x_2 = E(x|D_1), y_2 = E(y|D_1);\\
&\dots\\
&x_n = E(x|D_{n-1}), y_n = E(y|D_{n-1}).
\end{aligned}$$

It can be proved the estimation of (x_n y_n) has the best performance within these series of estimations

Choose (x_n y_n) as the Statistical Estimation

Statistical Estimation

- Rewrite it as two parts: one is certain part, the other is uncertain part

$$x_n = f_n(x_{n-1}, y_{n-1}, \cdots, x_1, y_1, x_0, y_0; s_{n1}, \cdots, s_{nm_n}; l_{n1}, \cdots, l_{nm_n}) + \epsilon_n,$$

$$y_n = g_n(x_{n-1}, y_{n-1}, \cdots, x_1, y_1, x_0, y_0; s_{n1}, \cdots, s_{nm_n}; l_{n1}, \cdots, l_{nm_n}) + \eta_n,$$

- By the iterative relationship, the Statistical Estimation can be rewrote as follow

$$x_n = f_n^*(x_0, y_0; s_{11}, \cdots, s_{1m_1}, \cdots, s_{n1}, \cdots, s_{nm_n}; l_{11}, \cdots, l_{1m_1}, \cdots, l_{n1}, \cdots, l_{nm_n}) + \epsilon_n^*,$$

$$y_n = g_n^*(x_0, y_0; s_{11}, \cdots, s_{1m_1}, \cdots, s_{n1}, \cdots, s_{nm_n}; l_{11}, \cdots, l_{1m_1}, \cdots, l_{n1}, \cdots, l_{nm_n}) + \eta_n^*,$$

- Find out the estimation as the iterative algorithm does

Two Special Structures

The Average Structure

Consider each snapshot has the same contribution to the estimation

$$x_n = \frac{1}{n}\sum_{i=1}^{n} x^i, y_n = \frac{1}{n}\sum_{i=1}^{n} y^i.$$

$$\begin{aligned} f_n &= (f^1 + f^2 + \cdots + f^n)/n, \\ g_n &= (g^1 + g^2 + \cdots + g^n)/n, \\ \epsilon_n &= (\epsilon^1 + \epsilon^2 + \cdots + \epsilon^n)/n, \\ \eta_n &= (\eta^1 + \eta^2 + \cdots + \eta^n)/n, \end{aligned}$$

Two Special Structures

The Special Weighted Structure

- The contribution to the estimation will be discounted one half with the time passes

$$
\begin{aligned}
&x_1 = x^1, y_1 = y^1, \\
&x_2 = (x^2 + x_1)/2, y_2 = (y^2 + y_2)/2\ , \\
&\cdots \\
&x_n = (x^n + x_{n-1})/2, y_n = (y^n + y_{n-1})/2
\end{aligned}
$$

where

$$
\begin{aligned}
&f_1 = f^1, g_1 = g^1, \epsilon_1 = \epsilon^1, \eta_1 = \eta^1; \\
&\cdots \\
&f_n = (f^n + f_{n-1})/2, g_n = (g^n + g_{n-1})/2, \epsilon_n = (\epsilon^n + \epsilon_{n-1})/2, \eta_n = (\eta^n + \eta_{n-1})/2.
\end{aligned}
$$

Current Findings

Lookup table

Three types models

Divide the antennas into three types, each type builds up an ellipse model

Region	e1	e2	e3	alpha	Region	e1	e2	e3	alpha
Aberdeen	0.3	0	0	6.4	ShamTseng	0.9	0.1	0	5.3
CauseWayBay	0.5	0.9	0	4.2	ShaTin	0.5	0.8	0	5.9
Central	0	0.9	0	6.5	ShekKipMeiPark	0.8	0	0	5.8
CheungShaWan	0	0.6	0	8.8	SheungShui	0	0	0	6.9
FoTan	0.9	0	0	6.3	SheungWan	0	0.9	0	7.1
HappyValley	0.7	0	0	6.8	TaiKooShing	0.9	0	0	5.4
HungHom	0.8	0	0	7	TaiWai	0.3	0	0	6.4
KowloonBay	0	0	0.4	8.7	TaiWoHau	0.2	0	0	5.8
KowloonCity	0.6	0.7	0	6.2	TinShuiWai	0.9	0	0	5.6
KowloonTong	0	0	0.4	7.2	ToLoHighway	0.1	0	0	5.4
KwaiFong	0.9	0.9	0	6.1	TsingYi	0.1	0	0	7
KwunTong	0	0	0	8.2	TsuenWan	0.9	0.9	0	8.9
LaiChiKok	0	0	0	7.8	TszWanShan	0.1	0	0	7.1
LaiKing	0.6	0.9	0.5	7.1	TuenMun	0.4	0	0.4	6.4
MaOnShan	0	0	0	8.9	WanChai	0.1	0.9	0.9	7
Mongkok	0.5	0.9	0.1	5.3	WongTaiSin	0.9	0	0	7.2
PE-MK	0.4	0.9	0.1	8.9	YauTong	0.1	0	0	6.2
PrinceEdward	0	0.9	0	7.9	YauYatChuen	0.3	0	0	7
ShamShuiPo	0.4	0.3	0.7	8.9	YuenLong	0.9	0	0	5.3

Result of the Geometric Algorithm

Give the special points values: Mean, Standard deviation, 67% and 90% point value

Region	Aver.	Std.	67%	90%	Region	Aver.	Std.	67%	90%
Aberdeen	240.55	146.23	247.33	490.56	ShamTseng	1907.11	845.49	2543.35	3027.96
CauseWayBay	256.83	189.26	324.81	651.46	ShaTin	347.48	170.48	395.34	565.30
Central	108.12	64.61	140.43	198.41	ShekKipMeiPark	320.92	186.32	366.48	584.90
CheungShaWan	119.86	67.91	156.51	213.10	SheungShui	492.11	307.42	803.50	888.64
FoTan	251.15	143.00	299.35	465.62	SheungWan	116.48	63.43	151.79	199.32
HappyValley	331.35	207.91	396.63	622.09	TaiKooShing	225.64	146.76	253.14	424.71
HungHom	931.83	530.23	1369.27	1598.5	TaiWai	227.45	90.65	268.78	322.31
KowloonBay	201.45	107.26	233.50	328.65	TaiWoHau	252.71	136.89	287.68	458.90
KowloonCity	220.21	127.16	272.06	503.10	TinShuiWai	375.17	180.10	487.58	641.71
KowloonTong	252.34	155.49	311.68	498.18	ToLoHighway	918.92	614.16	1375.36	2296.17
KwaiFong	187.20	110.77	209.55	347.50	TsingYi	482.61	242.95	587.93	838.96
KwunTong	113.86	60.43	133.78	217.37	TsuenWan	145.52	60.36	159.84	234.84
LaiChiKok	328.71	197.89	403.47	599.42	TszWanShan	246.83	128.49	292.24	441.80
LaiKing	783.46	313.36	976.17	1287.32	TuenMun	334.14	153.98	425.46	520.87
MaOnShan	356.42	122.84	404.56	486.20	WanChai	166.22	109.99	185.72	314.63
Mongkok	95.06	53.59	119.77	166.72	WongTaiSin	260.43	145.77	310.73	473.51
PE-MK	111.69	75.66	138.43	202.52	YauTong	458.96	323.61	537.43	834.18
PrinceEdward	120.89	73.05	136.09	224.27	YauYatChuen	226.48	111.04	274.05	381.47
ShamShuiPo	104.42	58.75	118.45	196.07	YuenLong	238.59	242.99	213.62	769.44

Result of the Iterative Algorithm

Give the special points values: Mean, Standard deviation, 67% and 90% point value

Region	Aver.	Std.	67%	90%	Region	Aver.	Std.	67%	90%
Aberdeen	233.21	140.20	248.73	491.38	ShamTseng	1896.60	855.40	2531.23	3022.95
CauseWayBay	247.97	188.23	307.71	614.57	ShaTin	318.89	160.89	409.60	614.93
Central	106.14	63.97	139.86	195.63	ShekKipMeiPark	310.25	186.34	358.65	574.62
CheungShaWan	120.55	68.59	156.68	212.46	SheungShui	492.11	307.42	803.50	888.64
FoTan	250.58	133.61	299.43	435.65	SheungWan	116.12	63.52	151.56	198.76
HappyValley	332.06	216.64	387.05	605.80	TaiKooShing	189.37	120.11	218.64	374.64
HungHom	930.25	531.46	1366.67	1598.50	TaiWai	225.82	90.62	268.21	319.17
KowloonBay	201.41	107.25	233.14	328.65	TaiWoHau	248.50	134.61	284.51	486.79
KowloonCity	221.56	129.57	259.86	504.56	TinShuiWai	375.94	195.22	490.20	655.78
KowloonTong	252.24	155.48	311.70	498.34	ToLoHighway	915.57	609.12	1352.25	2259.92
KwaiFong	187.06	103.85	233.06	329.67	TsingYi	484.75	240.36	586.58	838.97
KwunTong	113.86	60.43	133.78	217.37	TsuenWan	145.34	60.38	159.47	234.46
LaiChiKok	328.71	197.89	403.47	599.42	TszWanShan	240.37	126.59	288.68	433.32
LaiKing	776.12	308.10	962.79	1287.12	TuenMun	314.75	150.53	403.99	499.47
MaOnShan	356.42	122.84	404.56	486.20	WanChai	167.04	109.52	187.42	320.69
Mongkok	92.49	52.43	116.39	158.34	WongTaiSin	257.55	141.22	314.06	468.18
PE-MK	110.43	75.53	134.88	201.60	YauTong	452.45	320.34	521.41	820.10
PrinceEdward	120.60	73.07	136.07	225.67	YauYatChuen	226.42	108.00	274.67	378.60
ShamShuiPo	104.67	58.75	122.54	195.54	YuenLong	237.57	243.58	214.91	769.39

Compare with different algorithms

- GEPM and IGEPM have better accuracy than CG and CT do
- SE of GEPM and IGEPM has the best effect

Algorithm	Average	Std. of Error
CG	335.11	319.61
CT	321.62	222.63
GEPM	285.27	309.80
IGEPM	282.68	309.05
SE of GEPM	260.26	285.64
SE of IGEPM	257.94	288.53

Future work

- Extend the propagation model in to 3-D model
- Find out a filter to handle the problem of signal fluctuation

Q&A